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In [ ]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.datasets import load_iris
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import confusion_matrix, accuracy_score, classification_report
import warnings
warnings.filterwarnings("ignore")
%matplotlib inline

In [ ]: iris = load_iris()
iris.keys()

In [ ]: x = pd.DataFrame(iris['data'], columns=iris['feature_names'])
y = pd.DataFrame(iris['target'], columns=['target'])

In [ ]: x.head()

In [ ]: x.shape, y.shape

In [ ]: x.info()

In [ ]: y.info()

In [ ]: x.describe()

In [ ]: scaler = StandardScaler()
x = scaler.fit_transform(x.values)

In [ ]: x_train, x_test, y_train, y_test = train_test_split(x, y.values, test_size=0.3, random_state=42)

In [ ]: x_train.shape, x_test.shape, y_train.shape, y_test.shape

In [ ]: model = GaussianNB()

In [ ]: model.fit(x_train, y_train)

In [ ]: y_pred = model.predict(x_test)

In [ ]: cm = confusion_matrix(y_test, y_pred)
print(cm)

In [ ]: sns.heatmap(cm, annot=True, fmt='d')
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()

In [ ]: print(f"TP value is {cm[0,0]}")
print(f"TN value is {cm[1,1] + cm[2,2]}")

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print(f"FP value is {cm[0,1] + cm[0,2]}")  
print(f"FN value is {cm[1,0] + cm[2,0]}")
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In [ ]: print(f"Accuracy score is {accuracy_score(y_test, y_pred)}")
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In [ ]: print(f"Error rate is {1 - accuracy_score(y_test, y_pred)}")
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In [ ]: print(f"Precision score is {precision_score(y_test, y_pred, average='macro')}
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In [ ]: print(f"Recall score is {recall_score(y_test, y_pred, average='macro')}")
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In [ ]: print(classification_report(y_test, y_pred))
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In [ ]:
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